

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 If $f(\theta) = \min. (|2x - 7| + |x - 4| + |x - 2 - \sin \theta|)$, where $x, \theta \in \mathbb{R}$, then maximum value of $f(\theta)$ is
(A) 2 (B) 3 (C) 4 (D) 5
- Q.2 The number of real solution of the equation $\cos^{-1}x + 2\cos^{-1}x + 3\cos^{-1}x = 6\pi$ is
(A) 0 (B) 1 (C) 2 (D) infinitely many
- Q.3 If $g(x) = 2\sqrt{2} \sin x + \cos x$ then maximum value of $P(x) = \sqrt{g(x) - 2} + \sqrt{4 - g(x)}$ is
(A) 0 (B) 1 (C) 2 (D) 3
- Q.4 Domain of the function $f(x) = \frac{\sin^{-1} 2x}{\sqrt{\cos x + \frac{x^2}{2} - 1}}$ is $[a, b] - \{c\}$, then $(b - a + c)$ equals
(A) 0 (B) 1 (C) 2 (D) 3
- Q.5 Let $f: (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $f(x) = 2\tan^{-1}(2013^x) - \frac{\pi}{2}$, then
(A) $f(-3) + f(3) = 0$ (B) $f(-2) - f(2) = 0$
(C) $f(x)$ is a decreasing function (D) $f^2(5) - f^2(-5)$ is non zero
- Q.6 Let $f(x) = \sqrt{6 - 5^{\sin^{-1}x} - 5^{1 - \sin^{-1}x}}$ and $g(x) = \sqrt{5^{\sin^{-1}x}}$ then range of the function $h(x) = f(x).g(x)$ is $[a, b]$, then $(a + b)$ equals
(A) 1 (B) 3 (C) 2 (D) 26
- Q.7 If the equation $x^3 + px^2 + qx + 1 = 0$, ($p < q$) has only one real root x_0 , then value of $2 \tan^{-1}(\operatorname{cosec} x_0) + \tan^{-1}(2 \sin x_0 \sec^2 x_0)$ is
(A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) 0
- Q.8 If $x = \alpha$ satisfies the equation $\sin^{-1}x + \cos^{-1}x^2 + \frac{\pi}{2} = 0$, then the value of $\frac{\sec^{-1}\alpha - \tan^{-1}\alpha}{\cot^{-1}\alpha - \operatorname{cosec}^{-1}\alpha}$ is equal to
(A) 0 (B) 1 (C) -1 (D) π
- Q.9 If $f(x) = \sum_{r=0}^{\infty} |x| \left(\tan^{-1} \left(\tan \frac{1}{2^r} \right) \right)$, then number of solution(s) of the equation $f(x) + x^2 = 1$ is
(A) 0 (B) 1 (C) 2 (D) 3

- Q.10 The number of solutions of the equation $\sin^{-1}x = 2 \tan^{-1}x$, is
 (A) 1 (B) 2 (C) 3 (D) 4

[PARAGRAPH TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 & 12

Let a_r ($a_r > 0 \forall r \in \mathbb{N}$) be the r^{th} term of a G.P. such that $\sqrt{a_1 \cdot a_3 \cdot a_5 \cdot a_7} = (18a_1 + a_3)^2$ and $\sum_{r=1}^{\infty} \frac{1}{a_r} = \frac{3}{4}$.

- Q.11 The value of $\sum_{r=1}^{10} \tan \left(\sin^{-1} \left(\frac{a_{r+1}}{\sqrt{a_r^2 + a_{r+1}^2}} \right) \right)$ equals
 (A) 15 (B) 20 (C) 30 (D) 40
- Q.12 The value of $\sin^{-1} \left(\sin \left(\frac{a_2}{a_1} \right) \right) + \tan^{-1} (\tan(a_1))$ is equal to
 (A) -1 (B) 0 (C) 1 (D) 2π

Paragraph for question nos. 13 & 14

Consider, $f(x) = \frac{|3-2|x||}{1+|x|}$ and $g(x) = \frac{(2|x|-3)^2}{3}$.

Two more functions $h(x)$ and $k(x)$ are defined as

$$h(x) = \max. \{f(x), g(x)\}$$

$$k(x) = \min. \{f(x), g(x)\}$$

- Q.13 If $(x_i, h(x_i))$ ($i = 1, 2, 3, \dots, n$) are the points on the graph of $y = h(x)$ where no unique tangent can be drawn then $\sum_{i=1}^n |x_i|$ equals
 (A) 3 (B) 5 (C) 7 (D) 8
- Q.14 If the equation $k(x) = p$ has four solutions then number of integral values of p is(are)
 (A) 0 (B) 1 (C) 2 (D) 3

[MULTIPLE CORRECT CHOICE TYPE]

Q.15 & Q.16 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.15 If $f: \mathbb{R} - \{-1, k\} \rightarrow \mathbb{R} - \{\alpha, \beta\}$ is a bijective function defined by $f(x) = \frac{(2x-1)(2x^2-4px+p^3)}{(x+1)(x^2-p^2x+p^2)}$
 (where $p \geq 0$), then identify which of the following statement(s) is(are) correct?
 (A) If $k \in (-1, 1)$ then $\alpha + \beta = 2$ (B) If $k \in (-1, 1)$ then $\alpha + \beta = 6$
 (C) If $k \in (1, 3)$ then $\alpha + \beta = 4$ (D) If $k \in (1, 3)$ then $\alpha + \beta = 6$

- Q.16 If equation $\sin^{-1}\sqrt{x} + \cos^{-1}\sqrt{x^2-1} + \tan^{-1}\tan y = k$ has atleast one solution, then $k \in \left(\frac{p\pi}{2}, \frac{q\pi}{2}\right)$, where $p, q \in \mathbb{I}$, then value of $(p+q)$ is greater than or equal to
 (A) 2 (B) 3 (C) 4 (D) 5

[INTEGER TYPE]

Q.17 to Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

- Q.17 Let $f(x)$ be a real valued function such that $|f(x) + x^2 + 1| \geq |f(x)| + |x^2 + 1|$ and $f(x) \leq 0$, then find the absolute value of $\sum_{r=1}^5 (1 + f(r))$.
- Q.18 Let $f(x) = x^2 - 3x - 4$ and $g(x) = \operatorname{sgn} x$
 then sum of all the roots of the equation $g(f(x)) = \sin \frac{\pi x}{2}$ when $x \in [-10, 20]$
 (where sgn denotes signum function)
- Q.19 If complete domain of the function $f(x) = \frac{5 + 3x + \tan^{-1} x}{\log_{2013} \left(\frac{1}{2 - \sin 3x} \right)} + \sqrt{10 - x} + \sqrt{x}$ is $[a, b] - A$.
 Let L is the sum of all entries in set A , then $\frac{6L}{\pi} + (b - a)$ equals
- Q.20 If $\sum_{n=0}^{\infty} \tan^{-1} \left(\frac{\cot^{-1}(n^2 + 3n + 3)}{1 + \cot^{-1}(n+1) \cot^{-1}(n+2)} \right) = \tan^{-1} \left(\frac{p}{4} \right)$ then find the value of $[\cos^{-1}(\cos(p-1))]$.
 [Note: $[k]$ denotes greatest integer less than or equal to k .]

ANSWER KEY

Q.1	B	Q.2	B	Q.3	C	Q.4	B	Q.5	A
Q.6	C	Q.7	A	Q.8	B	Q.9	C	Q.10	C
Q.11	C	Q.12	A	Q.13	D	Q.14	B	Q.15	AD
Q.16	ABC	Q.17	[5]	Q.18	[0041]	Q.19	[0055]	Q.20	[2]

